SIMERJIT KAUR NATH

Singapore · LinkedIn · GitHub · Website · Google Scholar

MSc AI candidate at NTU (Jan 2027) who builds AI systems that hold up on real data. LLM applications and multi-agent pipelines, multimodal deep learning for city-scale digital twins at A*STAR, and efficient fine-tuning; first-author paper accepted at the SENTIRE Workshop, IEEE ICDM 2026. 2 years at NielsenIQ delivering research for global consumer brands and leading an 11-person team.

PUBLICATIONS

Nath, S. K., Cambria, E., Pen, H. *SenticLens: A Production Sentiment-Analysis Framework with Behavioral Auditing of LLM Affective Coherence.* SENTIRE Workshop at IEEE ICDM 2026 (ICDM is ranked CORE A*). Proceedings of the 2026 IEEE International Conference on Data Mining Workshops (ICDMW), IEEE; Scopus-indexed. **Accepted (in press).** Sentic Labcast (video)

EXPERIENCE

Technical Project Lead (part-time), Kenza Living (Kenza AI) *Dec 2025 – Apr 2026, Aug – Oct 2026*

- Led the end-to-end build of a **bilingual (English/Arabic) community-management platform** for residential communities: **React Native** mobile app, **Node.js/Prisma** backend and **PostgreSQL on AWS**, now **live on the App Store, Google Play and the web**.
- Built an **LLM assistant** that turns resident messages into amenity bookings and service tickets, with server-side guardrails for security (authentication, role checks, rate limiting) and for the booking use case (booking details from server data, explicit confirmation, ticket fallback) to reduce incorrect actions.

Graduate Researcher, SenticNet Lab, NTU *Aug 2025 – Present*

- Built **SenticLens**, an end-to-end sentiment and LLM evaluation framework ingesting live reviews from **three platforms** (Reddit, Apple App Store, Google Play), scoring polarity, intensity and emotion, with guardrails flagging inconsistent model outputs before results reach downstream analytics.
- Across **24,000 audited records**, showed models disagree with themselves **twice as often** on real reviews as on synthetic ones (38.16% vs 18.49%) and contradict themselves outright **about six times as often** (2.32% vs 0.37%), on every model and brand.

Research Intern, A*STAR Institute of Advanced Intelligence and Computing (IAIC) *Apr – Aug 2026*

Computational Sustainability

- Owned the **four-source** ingestion pipeline for urban climate observations and established the quality-control procedures the team now relies on.

Graduate Student Research Assistant, NTU School of Social Sciences (part-time) *Feb – Apr 2026, Aug – Oct 2026*

- Built a local, offline LLM workflow for confidential interview transcripts with automated alignment and scoring, using LLM-assisted extraction, HDBSCAN with UMAP and Ward-linkage clustering.

Research Collaborator, LLM-Augmented Financial Signal Analysis *Jan 2026 – Present*

- Direct a **multi-agent LLM pipeline** for financial event interpretation, guiding experimental design, temporal-leakage control, catalyst validation and pipeline integrity, and co-supervising IIT Bombay undergraduate researchers.

Research Associate → Research Analyst → Team Lead, NielsenIQ, Mumbai, India *Jul 2023 – Jul 2025*

- Led an **11-member** ARC Europe task team to **97% feedback receipt**, a feedback score of 2.8 out of 3 and zero performance escalations.
- Delivered concept analysis and market research for **100+ FMCG launches** globally, including Reckitt and Nestlé, at 100% reported accuracy and a 3.9/4.0 project rating.

SELECTED TECHNICAL PROJECTS

- **RAG grounding audit** (open source, github.com/simer05/rag-grounding-audit, 2026): tested **3 open-weight LLMs** on **500 HotpotQA questions** under 6 controlled evidence conditions. Reordering the same passages flipped **8–14%** of answers between right and wrong while accuracy stayed flat, and models answered without evidence **26–40%** of the time.
- **Agentbox**, open-source contributor (2026–present, github.com/savannahfeder/agentbox): desktop app that runs many Claude Code and Codex agents in parallel, each in its own git worktree. First contributor to bring **Windows support**: traced why agents could not start on Windows, then fixed how Claude Code and Codex launch, how tool servers and per-task folders work, and the in-app terminal; merged upstream by the maintainer.
- **AdaptLink, AI seller intelligence for Shopee** (Sea x OpenAI Codex Hackathon, Regional Finalist, team of 4, Jun 2026): came up with the product idea and built the seller-intelligence dashboard (TypeScript/Next.js), turning conversion, reviews, buyer questions, reply times and competitor prices into a product health score and up to **5 margin-safe actions**; designed a **6-agent** seller workflow with model routing.

EDUCATION

Master of Science in Artificial Intelligence *Aug 2025 – Jan 2027 (expected)*

Nanyang Technological University, College of Computing and Data Science, Singapore

Bachelor of Science in Computer Science *Jul 2020 – Jun 2023*

SIES College, University of Mumbai, India · CGPA 9.63/10 (WES 3.86/4.00) · ***Silver Medalist***

AWARDS, LEADERSHIP AND SERVICE

Regional Finalist, Sea x OpenAI Codex Hackathon (2026) · Top 6 Finalist, American Express GenAI Hackathon (2025) · 2nd Place, TE Connectivity x AWS Hackathon (2025) · Bravo! Superstar, NielsenIQ/ARC Excellence Awards · Vice Chair, Operations and Logistics, TEDxNTU 2026

TECHNICAL AND RESEARCH SKILLS

AI/ML: LLM evaluation and observability, guardrails, behavioural auditing, prompt engineering, supervised fine-tuning (SFT) with LoRA/QLoRA/PEFT, quantisation and 4-bit inference, deep learning, multimodal learning, NLP, sentiment and emotion analysis, diffusion models, physics-informed ML, time-series analysis, inference optimisation

ML systems: local and self-hosted deployment, multi-agent and agentic systems, LangGraph, sim-to-real, surrogate and reduced-order modelling for digital twins, experimental design, temporal-leakage control, responsible AI, reproducibility

AI coding tools: Claude Code, OpenAI Codex, Google Antigravity, Cursor, GitHub Copilot; hands-on with Claude, GPT, Gemini, DeepSeek, GLM (Z.ai), Grok and open-weight models

Tools: Python, TypeScript, React Native, Node.js, Next.js, PostgreSQL, PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, SQL, Git, Docker, AWS, Linux/HPC, Tableau, Power BI, Advanced Excel